

Sentiment Analysis of Twitter Data using NLP & Transformers

Explore how NLP and transformer models analyze Twitter sentiments.



Introduction to Sentiment Analysis

Definition

Identifying and categorizing opinions in text data.

Applications

- Brand
- monitoring
- Market research
- Political analysis

Challenges

- Sarcasm detection
- Context understanding
- Evolving language

Example

Track public reaction to product launches on Twitter.



Data Collection and Cleaning

Extract Data

Use Twitter API and libraries like Tweepy.

Remove Duplicates

Reduce redundancy by 15% for cleaner data.

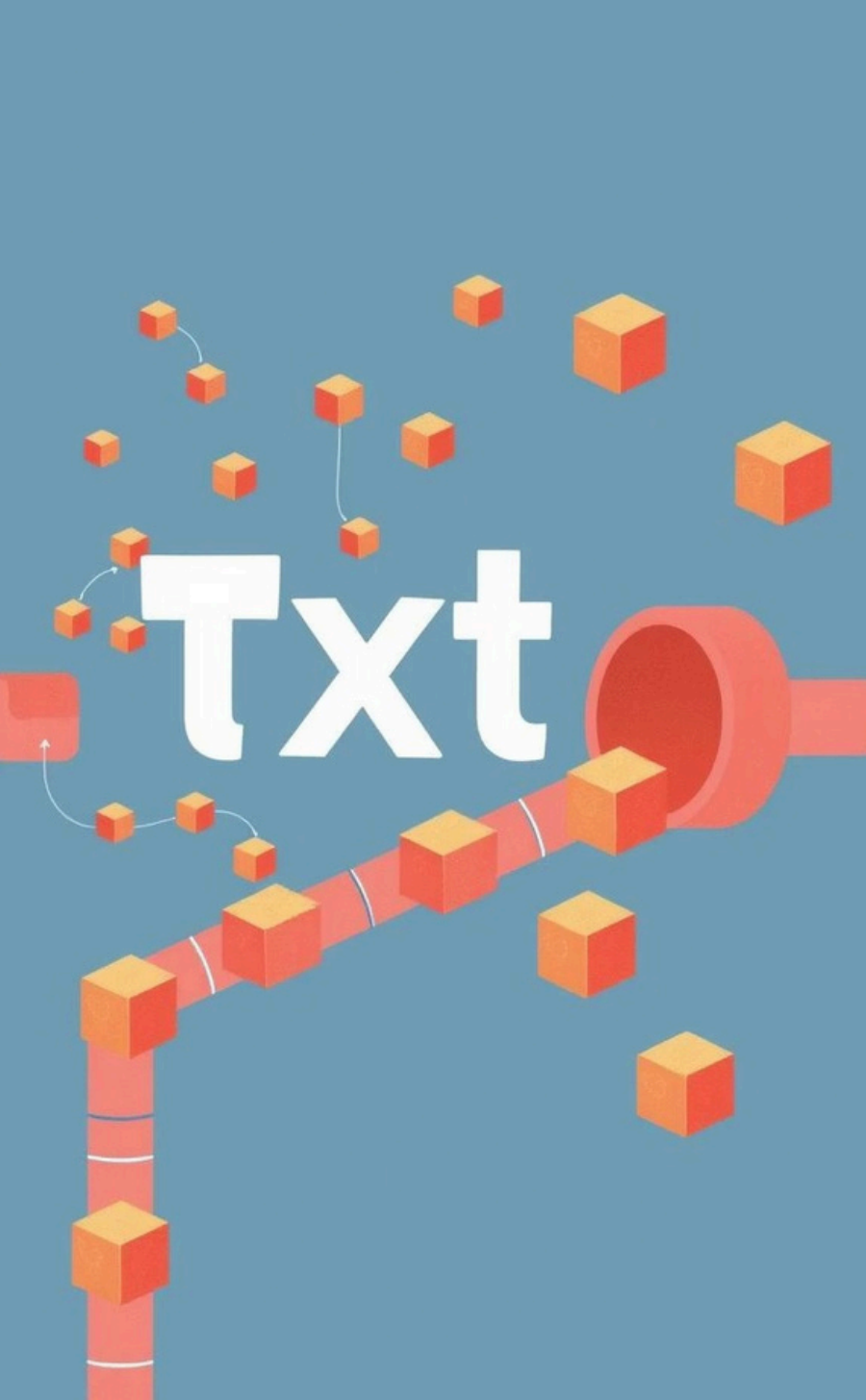
Handle Missing Values

Impute or remove; affects accuracy mildly by 5%.

Clean Text

Remove URLs, special characters; convert to lowercase.





Natural Language Processing (NLP) Techniques

1

Tokenization

Split text into words using NLTK or spaCy.

2

Stop Word Removal

Eliminate common words, reducing noise by 20%.

3

Stemming/Lemmatization

Normalize words to root forms, boosting accuracy by 10%.

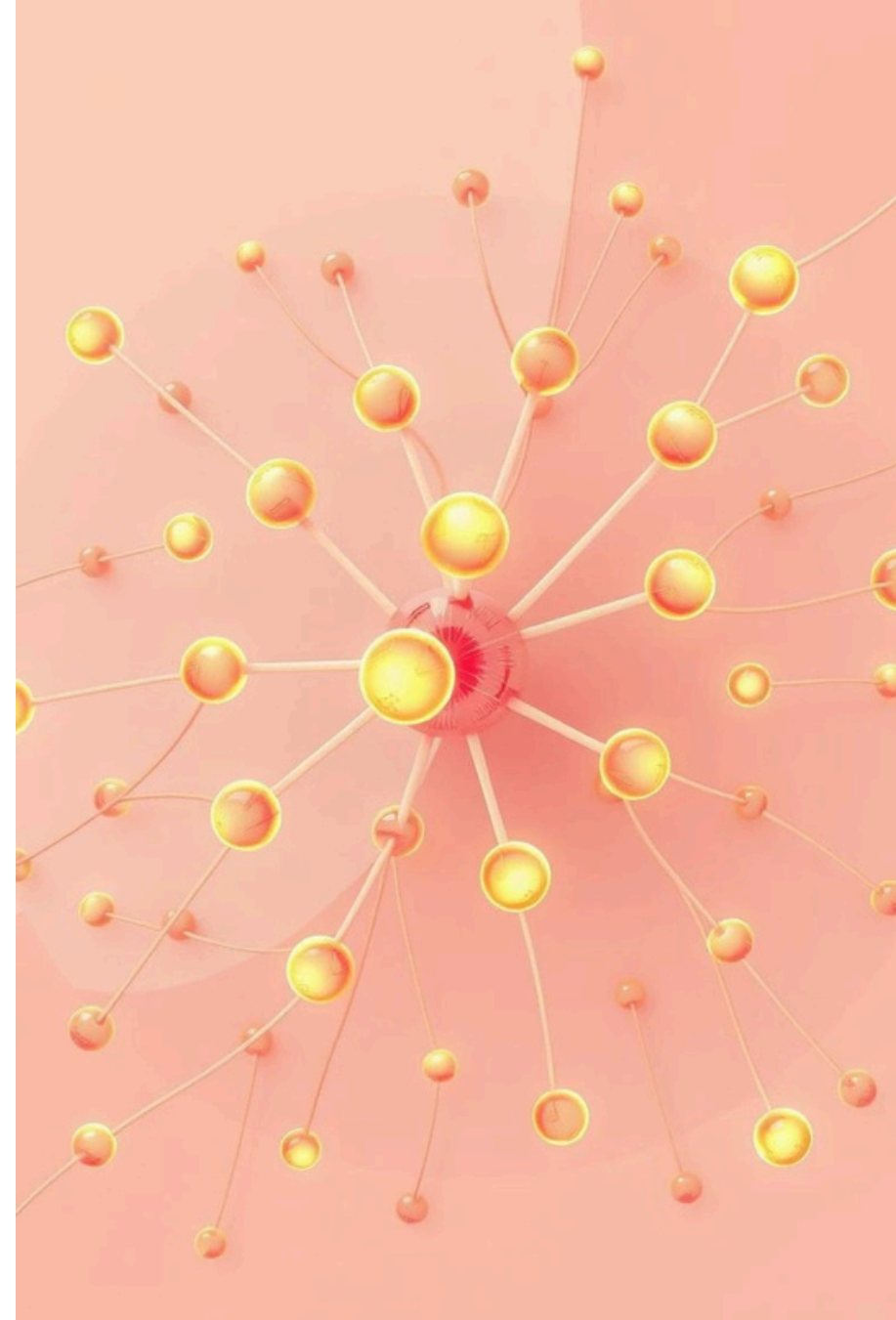
4

TF-IDF Vectorization

Assign weights based on word occurrences.

Transformer Models: BERT and Beyond

- ☐ **Transformer Architecture**
Revolutionizes NLP with attention mechanisms.
- ☐ **BERT**
Google's pre-trained bidirectional model for context understanding.
- ☐ **Fine-tuning**
Customize BERT for sentiment with 90%+ accuracy.
- ☐ **Variants**
RoBERTa and DistilBERT optimize performance and speed.



Model Training and Evaluation

Data Splitting

Train (80%), Validation (10%), Test (10%).

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

Cross-validation

Ensures robustness, prevents overfitting.

Performance Example

F1-score reaches 0.92 on testing data.

Results and Visualization



Sentiment Distribution

Positive, negative, and neutral sentiment breakdown.



Key Themes

Topic modeling reveals trends and discussions.



Time Series

Sentiment trends analyzed over time.



Geographic Insights

Mapping sentiment by location for regional analysis.



Conclusion and Future Directions

Summary

- ☐ Sentiment analysis reveals public opinion effectively.
- ☐ Limitations
Bias and data quality challenges persist.
- ☐ Future Work
 - Adopt advanced models like GPT-3
 - Enable real-time sentiment tracking
 - Expand to multilingual analysis

